Code No: 135BY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August/September - 2024
COMPUTER GRAPHICS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Name two advantage of LCD monitors over CRT monitors. [2]
b) What is the primary method used by the flood-fill algorithm to fill areas? [3]
c) What is the shear matrix for shearing an object along the x-axis by a factor of k? [2]
d) How does the viewing coordinate reference frame differ from the world coordinate system? [3]
e) What is an octree, and how does it partition 3-D space? [2]
f) How does perspective projection differ from orthographic projection? [3]
g) How does area subdivision improve visibility determination in 3-D graphics? [2]
h) What is the primary advantage of using flat shading over smooth shading techniques? [3]
i) Name one example of a computer animation language and its use. [2]
j) Why are timing and spacing important in animation? [3]

PART – B

(50 Marks)

2. Explain the differences between CRT and LCD monitors, including their advantages and disadvantages in the context of computer graphics. [10]

OR

3. Describe the mid-point ellipse drawing algorithm and compare it with the methods used for drawing circles. [10]

4. Analyze the impact of combining translation and scaling transformations on an object in 2-D space. How does the order of transformations affect the final result? [10]

OR

5. Describe the Sutherland-Hodgeman polygon clipping algorithm. How does it handle the clipping of polygons, and what are its strengths and limitations? [10]

6. Discuss B-spline curves and their advantages over other spline representations. How do B-splines offer greater control and flexibility in modeling complex shapes? [10]

OR

7. Evaluate the process of clipping in 3-D graphics. How are clipping algorithms used to remove parts of objects outside the view volume, and what challenges are associated with 3-D clipping? [10]

8. Explain how visible surface detection methods can be combined to optimize rendering performance. What are some common hybrid approaches, and how do they leverage the strengths of multiple techniques? [10]

OR

9. Describe the concept of texture mapping and its importance in surface rendering. How does texture mapping contribute to the realism of 3-D models? [10]

10. Explain the concept of motion specifications in computer animation. How are motion paths, speed, and acceleration defined and managed in animation systems? [10]

OR

11. Discuss the role of interpolation techniques in animation. How do interpolation methods like linear, spline, and Bézier interpolation affect the smoothness and realism of animated motions? [10]

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